

Mathematical Analysis V

Authors

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1 Intro

The aim of this GitHub repository is to delve deeper into biophysics, using simulated annealing and science data to humanize murine antibodies.

We use a frequentist analysis of the given data in `learn_human.txt` and `learn_mouse.txt` in order to measure entropies and correlations.

With those outputs we intend to fine-tune the simulated annealing algorithm for the humanization.

1.1 Raw data files

The `learn_human.txt` and `learn_mouse.txt` data entries are chains of 298 aminoacids each. Every aminoacid is represented by a character in this string: `-ACDEFGHIKLMNPQRSTVWY`. The `-` is a gap in the chain, as used in the notation convention.

1.2 Aminoacid's properties

Every aminoacid is a certain chemical compound. Thus, it has chemical properties, which vary from aminoacid to aminoacid. The used properties in this project are (in order): porperty-gap, hydrophobic, aromatic, aliphatic, polar, small, minuscule, positively charged, and negatively charged. Every aminacid has assigned at least one property, so for consistency we added the property version of the gap: property-gap, denoted as `-(PROP)`.

2 Representation and modelization of data

2.1 Aminoacids and properties

Every aminoacid will be coded as a 21-dimensional vector of real numbers between 0 and 1 yuxtaposed to a 9-dimensional vector of real numbers between 1 and 0. To interpret as follows:

- The first list will be the 21-dimensional vector, whose i -th component corresponds to the frequency of the i -th aminoacid from the forementioned aminoacid string.
- The second list will be the 9-dimensional vector, whose i -th component corresponds to the frequency of the i -th property from the forementioned property list.

As you can see, these variables are context dependent, since there is no reference for the frequency of aminoacids or properties. Therefore, if the chain is one that represents an actual protein, all frequencies may be 0 but one of them, which should be 1. Same will happen for the property vector. We will call this a *concrete aminoacid*.

Example: Let's encode the aminoacid cysteine (`C`). It is the third in the aminoacid list, thus, the first vector should be $v_{\text{elements}} = (0, 0, 1, 0, \dots, 0)$. Cysteine is hydrophobic and small, then: $v_{\text{properties}} = (0, 1, 0, 0, 0, 1, 0, 0, 0)$, since hydrophobic is in the 2nd position and small in the 6th.

Now changing the context, another usual setting is the so called *Schrödinger's aminoacid*. That will be the average aminoacid, which elements in both vectors will be *separately* the frequencies of every appearing aminoacid.

Separately means that the sum of the components of the first list is 1 and the sum of the components of the second list will also be 1.^[1]

2.2 Chains

A chain is an array of 298 aminoacids and it represents a whole antibody. Just as before, we will have concrete and Schrödinger's chains.

This is a $298 \times (21 + 9)$ matrix of split row conditioned frequencies.

¹This is true for every kind of aminoacid, whether it is concrete or Schrödinger's.